Lecture 4: Sep 10, 2026

Last time:

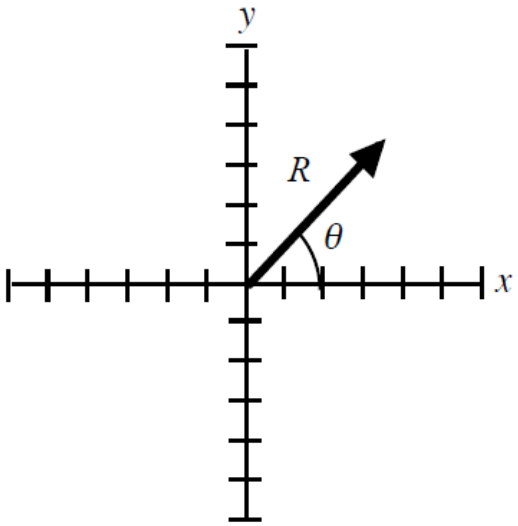
- Velocity and acceleration in SHM
- Adding SHMs together (same frequency, different amplitude)
- Review of complex math necessary for phasors

Today's class:

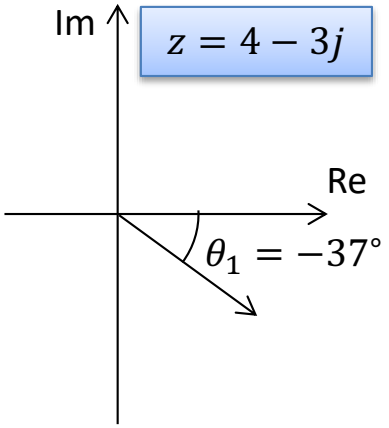
- Finish up review of complex math
- Phasor representation of SHM, using phasors
- Transverse vs longitudinal waves
- Travelling waves and the wave equation

- Complex numbers can be represented in two forms:
  - Rectangular: $z = x + jy$
  - Polar: $z = Re^{j\theta}$
- Use Euler's relation to convert polar → rectangular
- For rectangular → polar, use:

$$R = \sqrt{x^2 + y^2}$$
$$\theta = \tan^{-1}(y/x)$$



Inverse trig functions: a warning



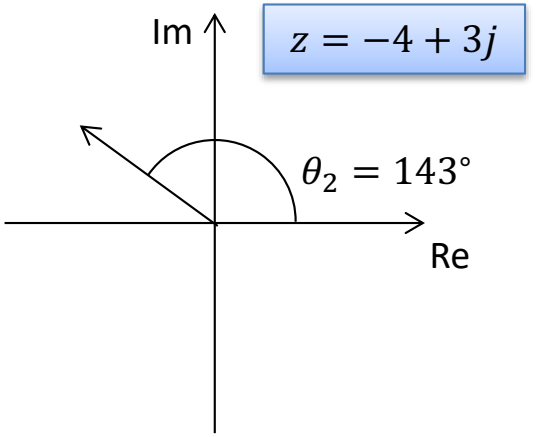
In both of these cases:

$$\frac{y}{x} = -\frac{3}{4}$$

so that

$$\tan^{-1}\left(\frac{y}{x}\right) = -37°$$

But that's not right for θ_2!



HINT: If in doubt, sketch it out.

- Your calculator will always return $\tan^{-1} x$ in the range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ [or in degrees: $(-90°, 90°)$].
  - This will be right about half the time.
- If necessary, add/subtract π $(180°)$

Top Hat Question

Express $z = -1 + 3j$ in polar notation $Ae^{j\phi}$.
What is the angle ϕ?

A. $108.4°$

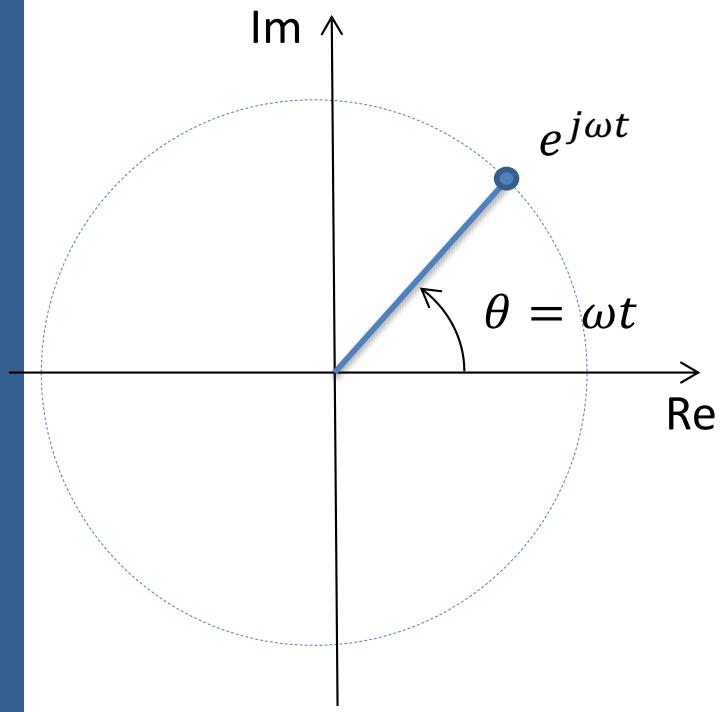
B. $71.6°$

C. $-71.6°$

D. $-108.4°$

PHASORS (PHASE VECTORS)

Phasor representation for SHM



Using Euler's identity

$$e^{j\theta} = \cos\theta + j\sin\theta$$

we see that

$$e^{j\omega t} = \cos\omega t + j\sin\omega t$$

describes a **rotating** unit vector in the complex plane.

(See animation at http://ophysics.com/w0.html)

$e^{j\theta} = \cos\theta + j\sin\theta$

Extending, SHM $x(t) = A\cos(\omega t + \phi_0)$

can be written as the real part of

$$\tilde{x}(t) = Ae^{j(\omega t+\phi_0)}$$

$$= \underbrace{Ae^{j\phi_0}}_{\text{Complex amplitude } \tilde{A}} \quad \underbrace{e^{j\omega t}}_{\text{Rotating unit vector}}$$

Complex amplitude $\tilde{A}$
- encodes amplitude and phase
- no dependence on time

Rotating unit vector
- time dependent

which is known as a **phasor** representation.

Addition with phasors

We can add

$$y = A_1 \cos(\omega t + \phi_1) + A_2 \cos(\omega t + \phi_2)$$

by adding the phasors:

$$y = \mathrm{Re}\left[A_1 e^{j(\omega t + \phi_1)} + A_2 e^{j(\omega t + \phi_2)}\right]$$

rather than plowing through a bunch of trig identities.

Adding SHM with phasors

Problem:

$$x_1(t) = A_1 \cos(\omega t + \phi_1)$$
$$x_2(t) = A_2 \cos(\omega t + \phi_2)$$
$$x_3(t) = x_1(t) + x_2(t)$$

Solution: Express in complex form:

$$x_1(t) = \mathrm{Re}\left[A_1 e^{j\phi_1} e^{j\omega t}\right] = \mathrm{Re}\left[\tilde{x}_1 e^{j\omega t}\right]$$

and add the complex amplitudes:

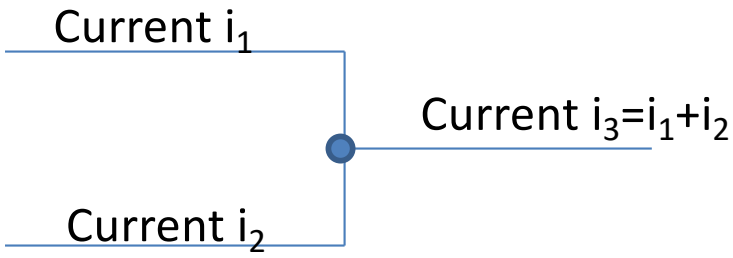
$$x_3(t) = \mathrm{Re}\left[\tilde{x}_1 e^{j\omega t}\right] + \mathrm{Re}\left[\tilde{x}_2 e^{j\omega t}\right]$$
$$= \mathrm{Re}\left[(\tilde{x}_1 + \tilde{x}_2)\, e^{j\omega t}\right]$$

5-MINUTE BREAK

Example

ex.2.1

Two alternating (AC) currents enter a 3-wire junction. The first is a current of 1.0 Amps at a phase of 30 degrees. The second is 2.0 Amps at a phase of 60 degrees. Find the resulting current out of the junction.



With phasors

First, let's write this out. Converting degrees to radians:

$$i_1 = 1.0\cos(\omega t + \pi/6)$$
$$i_2 = 2.0\cos(\omega t + \pi/3)$$

We want to find $i_3 = i_1 + i_2$.

The complex amplitudes are:

$$\tilde{i_1} = 1.0e^{j\cdot\pi/6}$$
$$\tilde{i_2} = 2.0e^{j\cdot\pi/3}$$

Recall: Euler's formula
$e^{j\theta} = \cos\theta + j\sin\theta$

To do the addition, switch to rectangular coordinates:

$$\tilde{i}_1 = 1.0e^{j\cdot\pi/6} = 1.0\cos\frac{\pi}{6} + 1.0j\sin\frac{\pi}{6}$$
$$= 0.8660 + 0.5j$$
$$\tilde{i}_2 = 2.0e^{j\cdot\pi/3} = 2.0\cos\frac{\pi}{3} + 2.0j\sin\frac{\pi}{3}$$
$$= 1.0 + 1.7321j$$

and then just add:

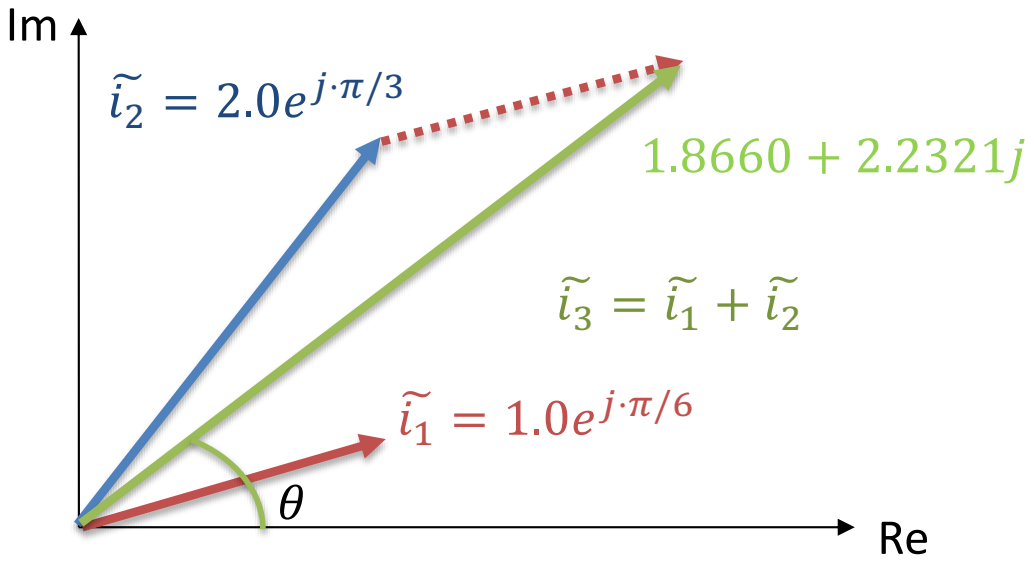
$$\tilde{i}_3 = \tilde{i}_1 + \tilde{i}_2 = 1.8660 + 2.2321j$$

To express the results in standard form, switch back to polar form:

$$\tilde{i}_3 = 1.8660 + 2.2321j$$

$$\left|\tilde{i}_3\right| = \sqrt{1.8660^2 + 2.2321^2} = 2.91$$

$$\theta = \tan^{-1}(2.2321/1.8660) = 50.1^o$$

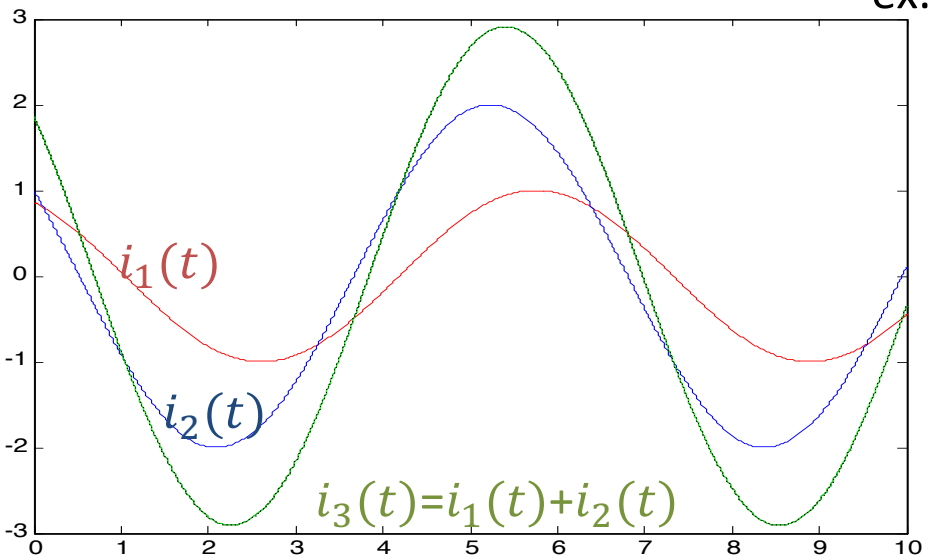


$$i_3(t) = \tilde{i}_3 e^{j\omega t}$$
$$= \left|\tilde{i}_3\right| e^{\theta} e^{j\omega t}$$
$$= 2.91\cos(\omega t + 50.1^o)$$

The resulting current is 2.91 Amps at a phase of 50.1 degrees

Verify

ex.2.1



Matlab:

```
>> wt=0:0.001:10;
>> i1=1.0*cos(wt+30*(pi/180));
>> i2=2.0*cos(wt+60*(pi/180));
>> i3=2.91*cos(wt+50.1*(pi/180));
>> plot(wt,i1,'r')
>> hold on
>> plot(wt,i2,'b')
>> plot(wt,i1+i2,'g')
>> plot(wt,i3,'k:','LineWidth',2)
```

Recipe for phasor addition

extra info

1. Write all sinusoidal motion in the form $A\cos(\omega t + \phi_0)$. (Don't use sines!)
2. Convert to phasor form $Ae^{j\phi_0}e^{j\omega t} = \tilde{A}e^{j\omega t}$
3. Add the complex amplitudes $\tilde{A}_1 + \tilde{A}_2$ in rectangular form
4. Convert the result to polar form
   - Careful of the multi-valued arctan()
5. Done: take the real part.

Beyond addition

- Concept can be extended to a variety of physical systems.
- Very useful for waves and signal analysis
- Differentiation, integration are well-defined and relatively straightforward:

$$\frac{d}{dt}\tilde{A}e^{j\omega t} = \left(j\omega\tilde{A}\right)e^{j\omega t}$$

- Wide application